

The empirical recurrence for OEIS A251831, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A251831 records “Empirical recurrence of order 65” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 107$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 68$. Its last coefficient is $c_{65} = 96 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A251831 is “Number of $(n+2) \times (2+2)$ 0..3 arrays with no row, column, diagonal or antidiagonal in any 3×3 subblock summing to 3 4 5 or 6.”. It has offset 1 and begins

3354, 27554, 262404, 2298772, 20314774, 182241310, 1624862992, 14480032642, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:33:59, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 107$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 214$ exact terms from $n = 3$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	4	c_{18}	−6283124	c_{35}	−58928040	c_{52}	1759938
c_2	29	c_{19}	11830444	c_{36}	−52615026	c_{53}	1135404
c_3	165	c_{20}	−16009987	c_{37}	47872481	c_{54}	−1230635
c_4	−64	c_{21}	25946091	c_{38}	−60007557	c_{55}	1038442
c_5	−1430	c_{22}	−8621699	c_{39}	−19117280	c_{56}	−450822
c_6	−5232	c_{23}	3071833	c_{40}	83824303	c_{57}	72278
c_7	969	c_{24}	27924555	c_{41}	−75005104	c_{58}	12394
c_8	13483	c_{25}	−54387954	c_{42}	65269776	c_{59}	−43011
c_9	3328	c_{26}	25988588	c_{43}	10745759	c_{60}	28963
c_{10}	15317	c_{27}	−40006966	c_{44}	−67134847	c_{61}	−8182
c_{11}	9745	c_{28}	−7119548	c_{45}	75286422	c_{62}	1708
c_{12}	336218	c_{29}	5567611	c_{46}	−65448484	c_{63}	8
c_{13}	−201477	c_{30}	39133265	c_{47}	19444017	c_{64}	−304
c_{14}	−54764	c_{31}	−74716756	c_{48}	17812434	c_{65}	96
c_{15}	−932015	c_{32}	172550421	c_{49}	−22991743		
c_{16}	−1225980	c_{33}	−155823544	c_{50}	18075117		
c_{17}	1602899	c_{34}	141849099	c_{51}	−9819682		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 68$, and it is the recurrence OEIS A251831 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{64} - \dots - c_{65}$. Its constant term is $-c_{65} = -96$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 96.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251831.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.